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Paper Code : CS-504D

PUID : 05017 (To be mentioned in the main answer script)

OBJECT ORIENTED PROGRAMMING

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

i) Dynamic method dispatcher is used for

a) resolving method overriding

b) resolving multilevel inheritance

c) resolving multiple inheritance anomally

d) none of these.

ii) What is the output of this program?

```
class box
{
int width;
int height;
int length;
int volume;
void volume()
{
volume = width*height*length;
}
}
class Output
{
public static void main(String args[])
{
box obj = new box();
obj.height = 1;
obj.length = 5;
obj.width = 5;
obj.volume();
System.out.println(obj.volume);
}
}
```

a) 0 b) 1

c) 25 d) 26.

iii) Which keyword is used by method to refer to the object that invoked it ?

a) Import b) Catch

c) Abstract d) This.

iv) Abstract class cannot have a constructor.

a) True

b) False.

v) Which operator is used to implement unsigned right shift of an integer ?

a) << b) >>

c) <<< d) >>>.

vi) Runnable is a

a) Class b) Method

c) Variable d) Interface.

vii) The ..... class is used to read characters from the file:

a) StreamReader b) CharacterReader

c) InputReader d) FileReader.

viii) DataInputStream class contains methods to read .....

a) Non primitive data type

b) Primitive data type

c) Formatted data

d) All of the above.

ix) Which of these keywords is not a part of exception handling ?

a) Try b) Finally

c) Thrown d) Catch.

x) Which of the following is true in the case of abstract class ?

a) Abstract constructors cannot be created

b) Abstract classes can not be inherited

c) An abstract class contains only abstract methods

d) All of these.

xi) What is the result of the following code

```
public class Outer
{
    private static int data = 10;
    private static int LocalClass()
    {
        class Inner
        {
            public int data = 20;
            private int getData()
            {
                return data;
            }
        };
        Inner inner = new Inner();
        return inner.getData();
    }
    public static void main(String[] args)
    {
```

```
        System.out.println(data * LocalClass());
    }
}
```

a) Execute with no output

b) Compilation Error

c) 200

d) None of these.

xii) What does AWT stand for ?

a) All Window Tools

b) Abstract Window Toolkit

c) All Writing Tools

d) None of these.

GROUP – B
(Short Answer Type Questions)

Answer any *three* of the following. 3 × 5 = 15

2. What is byte code ? What does the JVM do? Why Java is called platform independent language ? 1 + 2 + 2

3. Explain "public static void main(String args[])".

4. What is Final Keyword in Java? Give an example. 4 + 1

5. What's the difference between comparison done by equals method and == operator?

6. Write a program in java to explain the working 'this'.

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. Explain with the help of an example, how java gets benefited by using Interface ? How can you create your own package and add classes in that ? Explain with the help of an example. 6 + 6 + 3

8. What is the difference between Exception and Error ? Show with an example. What is the difference between throw and throws? Write program to handle different types of exception using multiple catch statement. When do we use finally block ? 3 + 1 + 4 + 5 + 2

9. Differentiate between Object Oriented Programming and Procedure Oriented Programming. Explain JDK, JVM and JRE? Write down the compilation and interpretation process of java programming using a flow chart. 3 + (2 + 2 + 2) + (3 + 3)

10. Define an interface perimeter using java that contains a method to calculate the perimeter of an object. Define two classes circle and Rectangle with suitable fields and

methods. Implement the interface "perimeter" in these classes. Write the appropriate main() method to create object of each class and test all the methods.

$3 + (4 + 4) + 4$

11. Write short notes on any *three* of the following giving suitable examples:

a) Standard Java Packages

b) Java AWT controls

c) NetBean

d) Java class loaders

e) JDBC models.